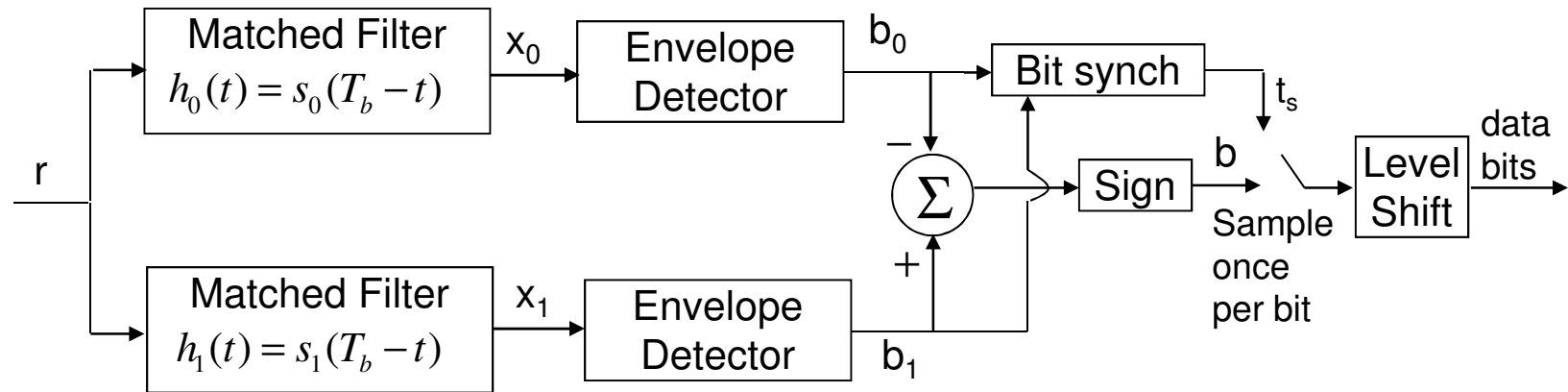


Non-coherent FSK Receiver Implementation in MATLAB

- Tx sends: $s(t) = \begin{cases} s_0(t) = A \sin(2\pi f_0 t), & \text{if "0"} \\ s_1(t) = A \sin(2\pi f_1 t), & \text{if "1"} \end{cases}, \quad 0 \leq t \leq T_b$

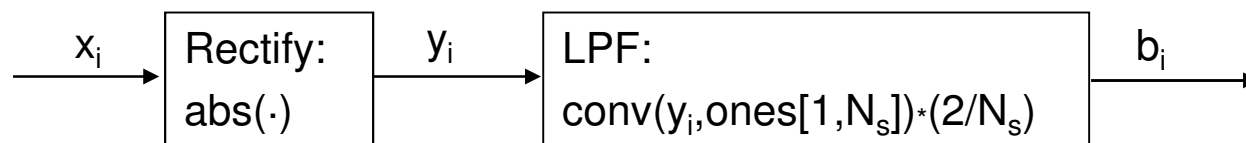
- Non-coherent FSK Receiver Block Diagram:



- Note: $s_0(T_b - t) = A \sin(2\pi f_0 (T_b - t)) = A \sin(2\pi \frac{n}{T_b} T_b - 2\pi f_0 t) = -A \sin(2\pi f_0 t)$
- Similarly, $s_1(T_b - t) = -A \sin(2\pi f_1 t)$

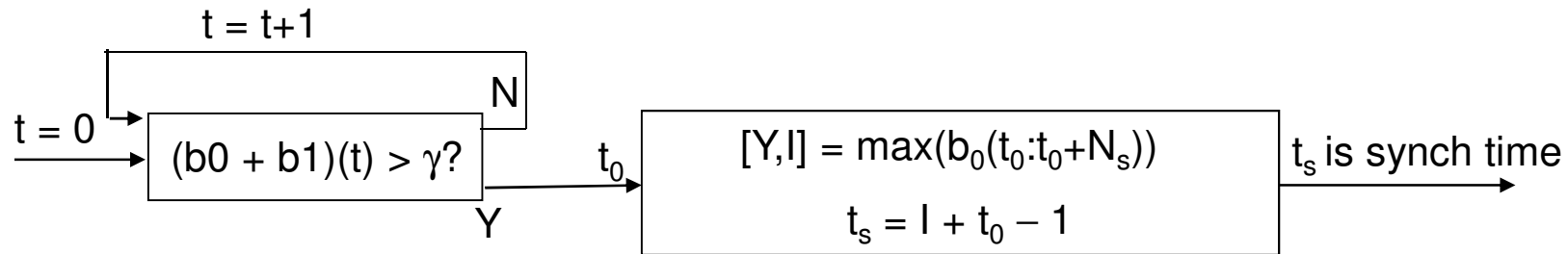
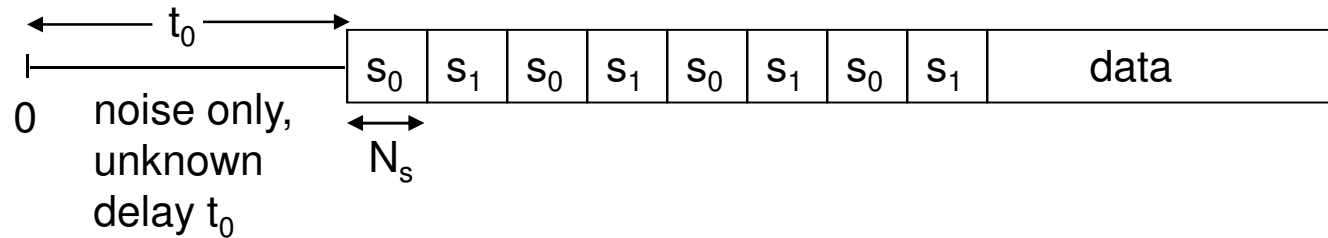
Matched Filter and Envelope Detector

- To compute matched filter outputs, use $x_i = \text{conv}(r, s_i) * (2/N_s)$, where $i = 0, 1$
- r is the received sample vector $r = [r_0, r_1, \dots, r_N]$, where $r_k = r(kT_s)$
- T_s is the sampling interval: $T_s = T_b/N_s$, where N_s is the # samples per bit.
- $s_i = [s_{i1}, s_{i2}, \dots, s_{iN_s}]$ is the sample vector, with $s_{ik} = s_i(kT_s)$, over one bit interval.
- The envelope detectors can be implemented as follows:



Bit Synchronization

- The received signal has a synchronization preamble, used for bit synch



- Sample the synch and data bits at times $t_s + i \cdot N_s$
- Threshold γ is set experimentally by plotting $b_0 + b_1$.
- l output of the max function is the index of the maximum.